FLATNESS OF THE TENSOR SQUARE AND WEAK ETALENESS DESCEND ALONG A FAITHFULLY FLAT EXTENSION

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement span contains the author's two-part enumerated list, delivered verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers; both point to the same descent lemma and are labelled with the same target.

Allowed standard prerequisites (editorial).

- The general descent lemma for flatness: flatness descends along a faithfully flat ring map. invoked at proof lines 30397-30398 and 30400-30401. Is the tool converting flatness upstairs into flatness downstairs in both parts.
- Flatness of the base change of a flat map, and flatness of a tensor product of flat modules. used at proof lines 30394-30395. Gives the flatness of the vertical arrows and the top horizontal arrow of the square.
- Weak etaleness as the conjunction of flatness of the map and flatness of the tensor square morphism. used at proof lines 30380-30381 and 30399-30401. Translates the two flatness statements into the weak etaleness conclusion.

Source reference. Original lemma 0584: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6767–6767. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $A \to B \to C$ be ring maps.*

(1) *If $B \to C$ is faithfully flat and $C \otimes_A C \to C$ is flat, then $B \otimes_A B \to B$ is flat.*
(2) *If $B \to C$ is faithfully flat and $A \to C$ is weakly étale, then $A \to B$ is weakly étale.*

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Proof. Assume $B \to C$ is faithfully flat and $C \otimes_A C \to C$ is flat. Consider the commutative diagram

$$\begin{array}{ccc} C \otimes_A C & \longrightarrow & C \\ \uparrow & & \uparrow \\ B \otimes_A B & \longrightarrow & B \end{array}$$

The vertical arrows are flat, the top horizontal arrow is flat. Hence C is flat as a $B \otimes_A B$-module. The map $B \to C$ is faithfully flat and $C = B \otimes_B C$. Hence B is flat as a $B \otimes_A B$-module by Algebra, Lemma 0584. This proves (1). Part (2) follows from (1) and the fact that $A \to B$ is flat if $A \to C$ is flat and $B \to C$ is faithfully flat (Algebra, Lemma 0584). $\square$